

ME 103

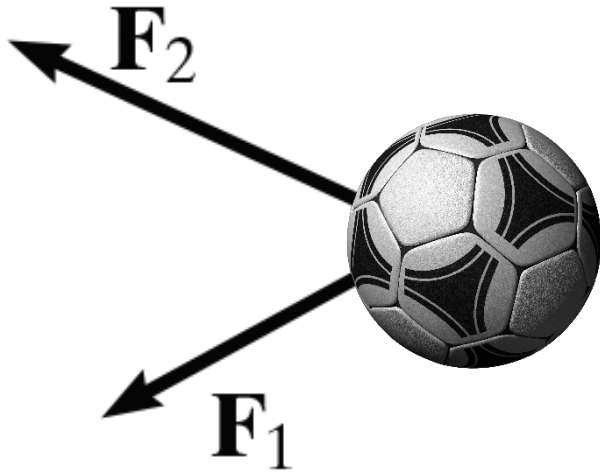
Lecture 11

Kinetics of **Particles**

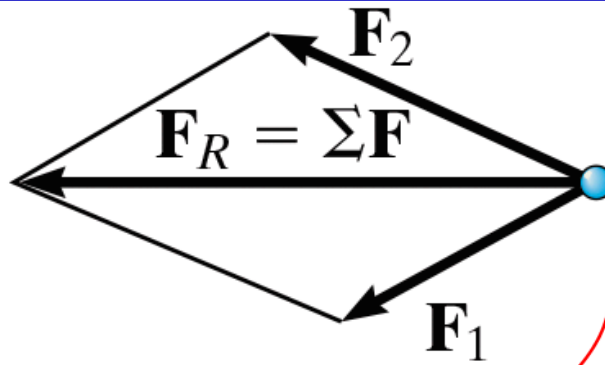
$$\vec{F} = m\vec{a}$$



Diagrams

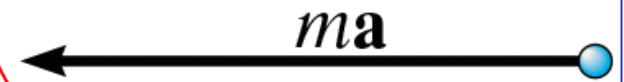


$$\Sigma \vec{F} = m\vec{a}$$



Free-body
diagram

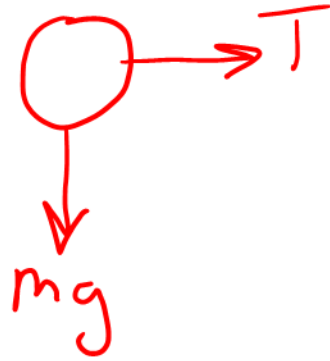
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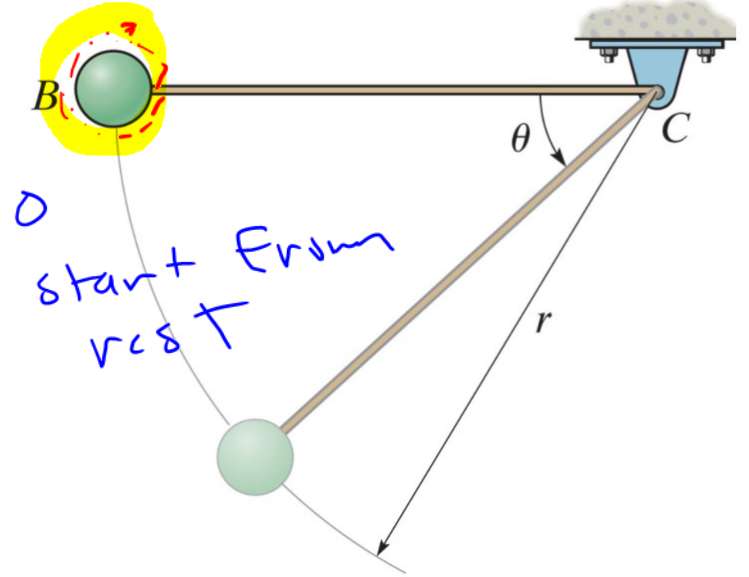
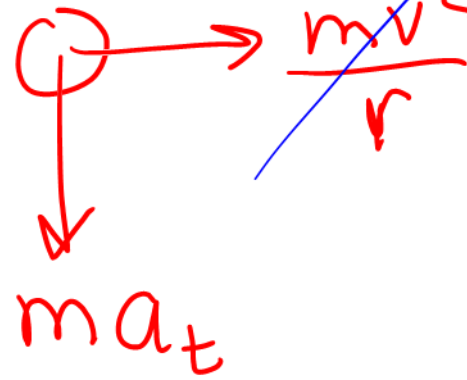
Kinetic
diagram

The 5kg pendulum bob is released from rest when $\theta = 0$. Determine the initial tension in the cord. Also determine the tension at the instant $\theta = 45^\circ$. Neglect the size of the bob and assume $r = 2\text{m}$

FBD

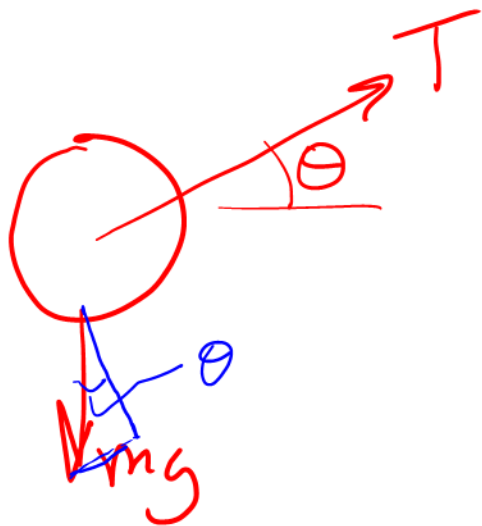


KV

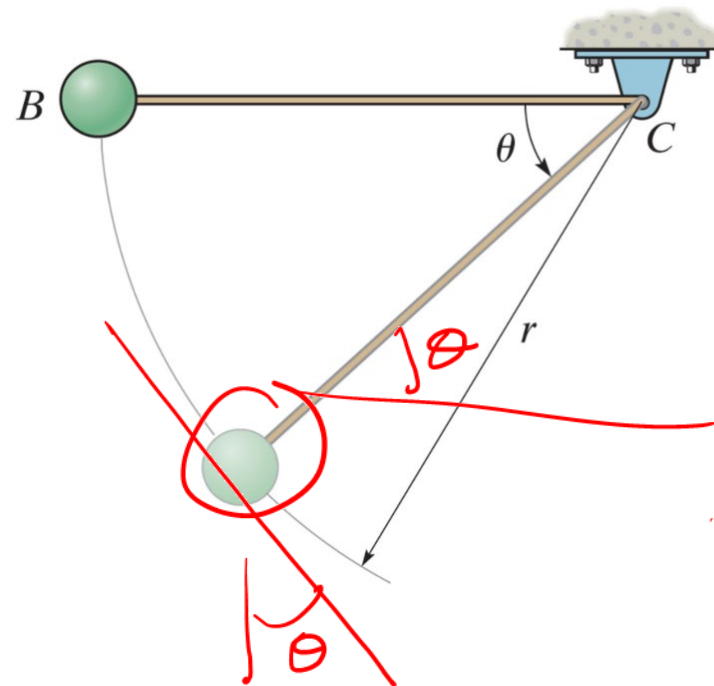
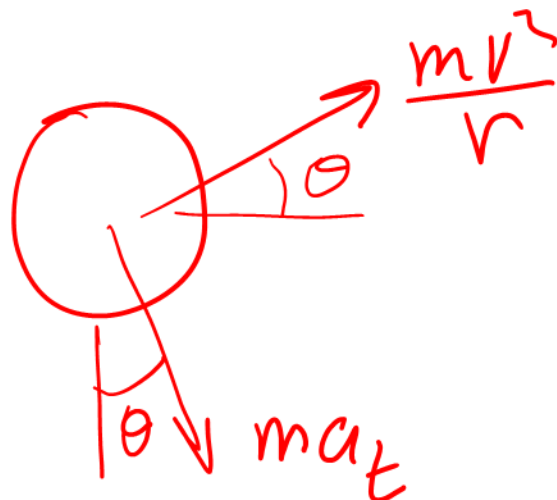


$$\begin{aligned} \rightarrow \Sigma F_x &= ma_x \\ T &= 0 \end{aligned}$$

FBD



KD



$$+ \rightarrow \Sigma F_n = ma_n$$

$$T - mg \sin \theta = \frac{mv^2}{r}$$

$$+ \searrow \Sigma F_t = ma_t$$

$$\cancel{mg \cos \theta} = \cancel{ma_t}$$

$$a_t = g \cos \theta$$

integrate a_t to get V

but a_t is not constant

$$a_t = g \cos \theta$$

For a circular path
 $ds = r d\theta$

$$a ds = v dv$$

~~$$a = \frac{dv}{dt}$$~~

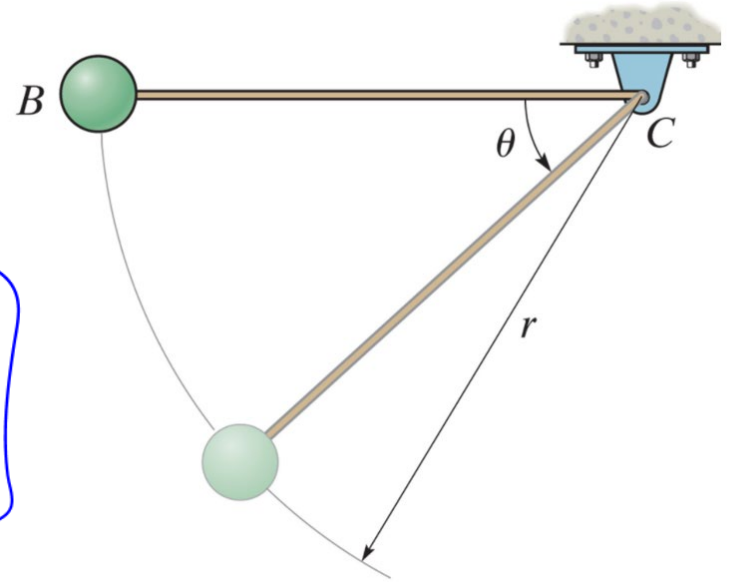
~~$$v = \frac{dx}{dt}$$~~

$$\int a r d\theta = \int v dv$$

$$\int_0^{\pi/4} (g \cos \theta) r d\theta = \int v dv$$

$r = 2m$

$$\begin{aligned} & r g \sin \theta \Big|_0^{\pi/4} \\ &= \frac{v^2}{2} \Big|_0^v \end{aligned}$$



$$\frac{1}{2}V^2 = 13.87 \Rightarrow \underbrace{V = 5.63 \text{ m/s}}$$

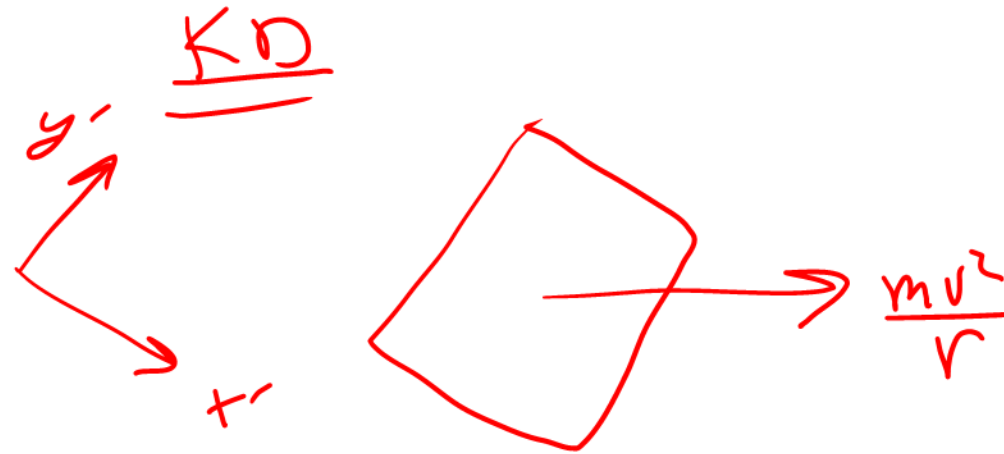
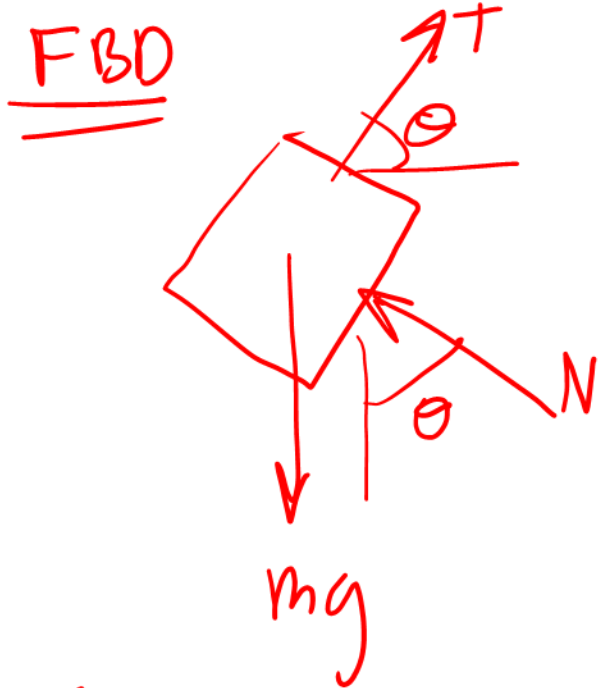
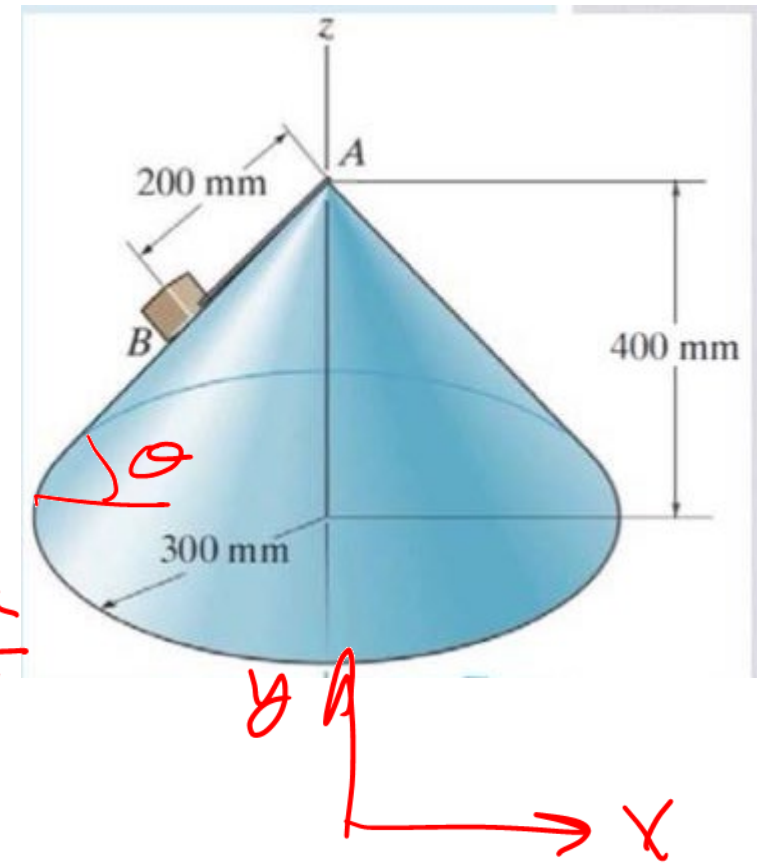
$$T - mg \sin \theta = \frac{mv^2}{r}$$

$$T = mg \sin \theta + \frac{mv^2}{r}$$

$$\uparrow$$
$$\theta = 45$$

$$\Rightarrow T = 104 \text{ N}$$

The smooth block b having a mass of 0.2 kg is attached to the vertex A of the right circular cone using a light cord. The cone is rotating at constant angular speed about the z-axis such that the block attains a speed of 0.5 m/s. At this speed, determine the tension in the cord and the reaction the cone exerts on the block. Neglect the size of the block.

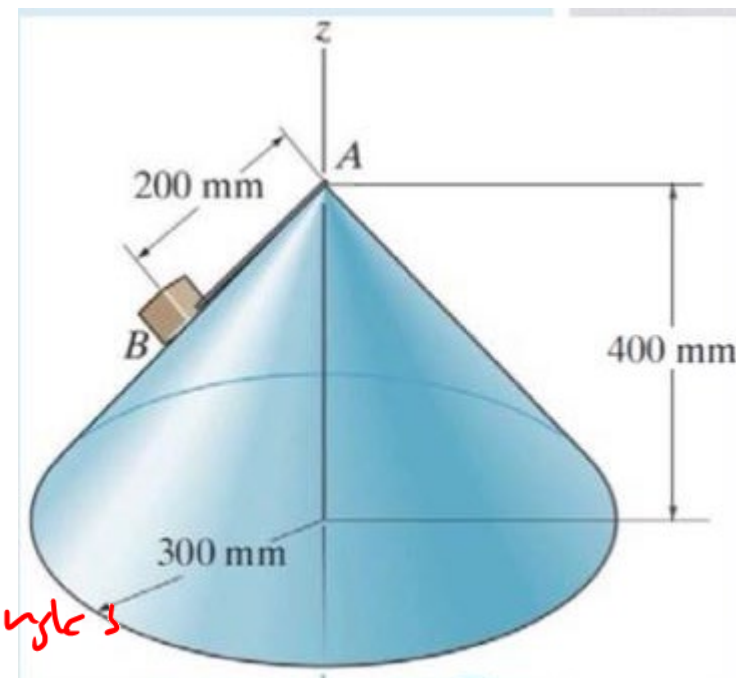
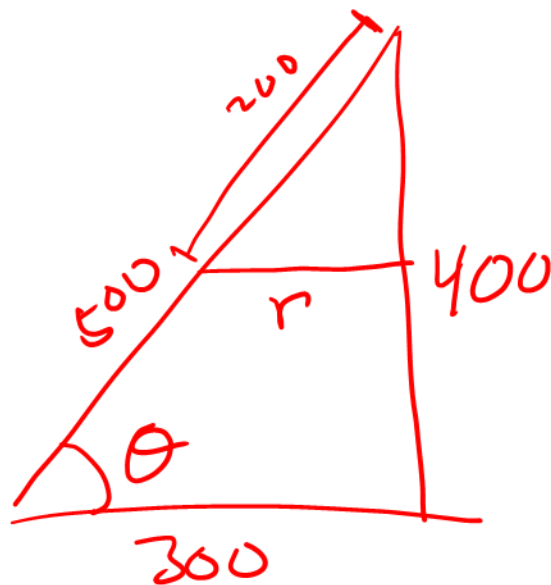


$$+\uparrow \sum F_y = 0$$

$$-mg + N \cos \theta + T \sin \theta = 0$$

$$+\rightarrow \sum F_x = \frac{mv^2}{r}$$

$$T \cos \theta - N \sin \theta = \frac{mv^2}{r}$$



similar triangles

$$\cos \theta = \frac{300}{500} = \frac{3}{5}$$

$$\sin \theta = \frac{400}{500} = \frac{4}{5}$$

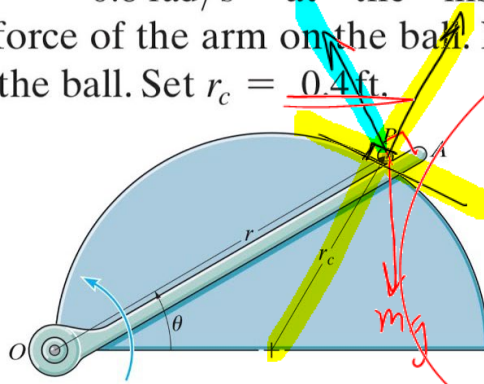
$$\frac{r}{200} = \frac{300}{500}$$

$$r = 200 \left(\frac{3}{5} \right) = \underline{\underline{120 \text{ mm}}}$$

$$T = 1.82 \text{ N}$$

$$N = 0.844 \text{ N}$$

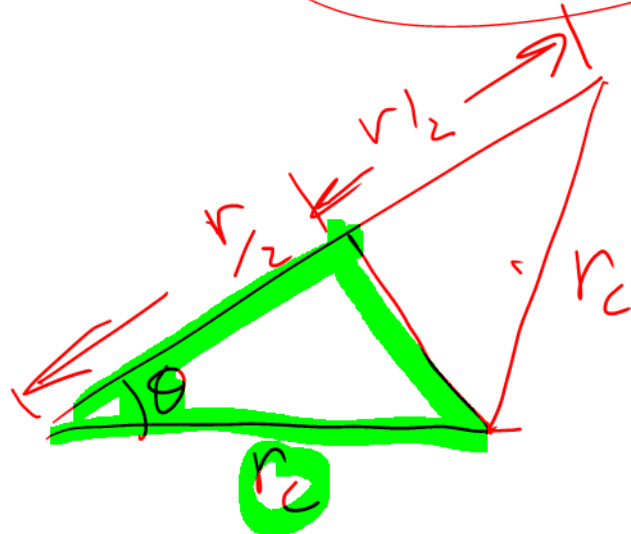
***13-100.** The 0.5-lb ball is guided along the vertical circular path $r = 2r_c \cos \theta$ using the arm OA . If the arm has an angular velocity $\dot{\theta} = 0.4 \text{ rad/s}$ and an angular acceleration $\ddot{\theta} = 0.8 \text{ rad/s}^2$ at the instant $\theta = 30^\circ$, determine the force of the arm on the ball. Neglect friction and the size of the ball. Set $r_c = 0.4 \text{ ft}$.



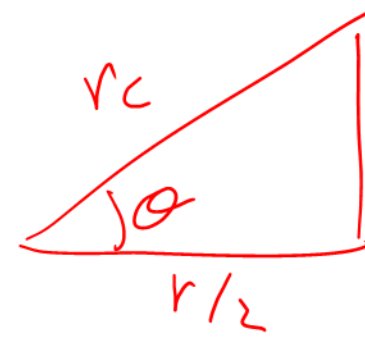
$$\theta = 30^\circ$$

$$\dot{\theta} = 0.4 \frac{\text{rad}}{\text{s}}$$

$$\ddot{\theta} = 0.8 \frac{\text{rad}}{\text{s}^2}$$



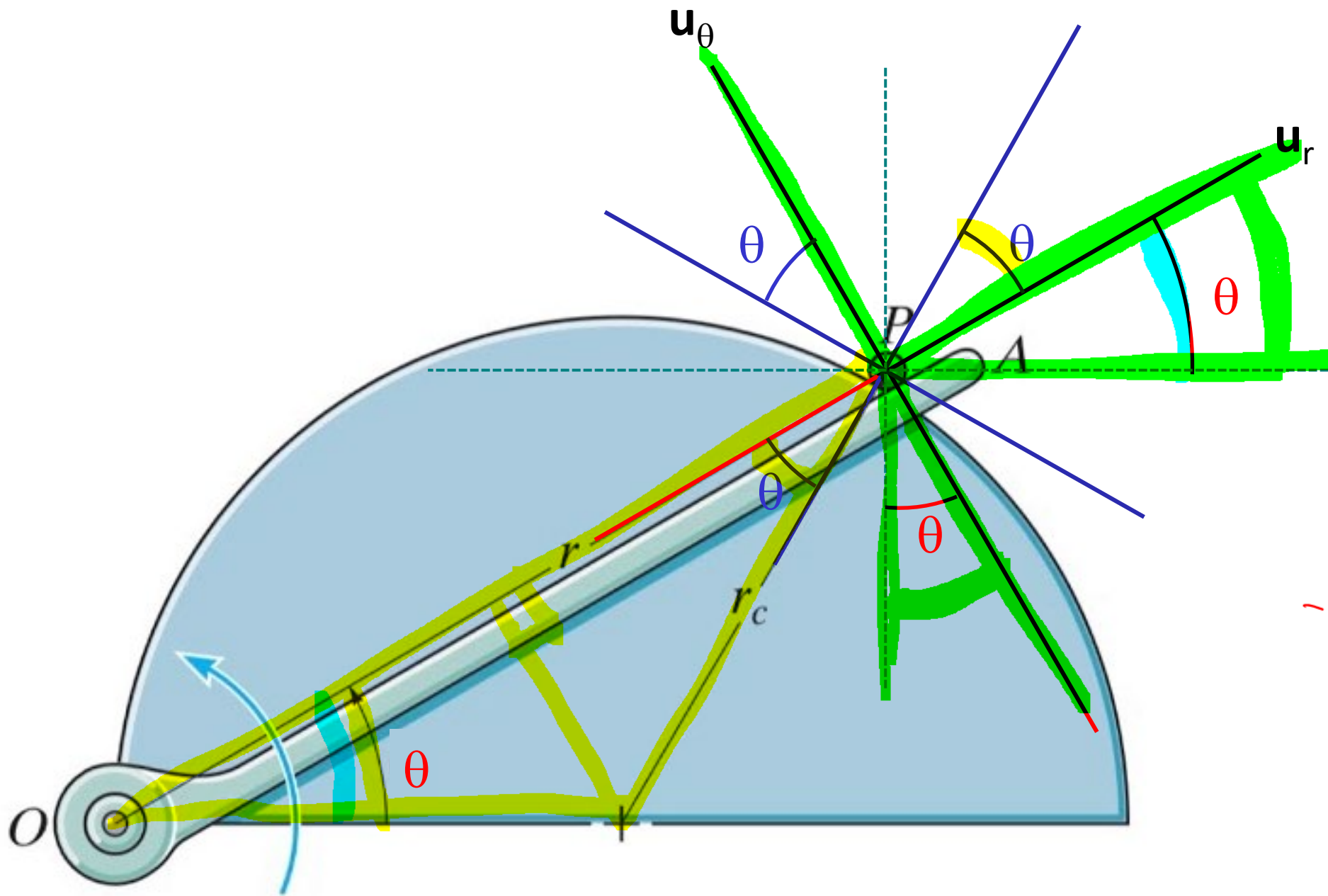
$$r = 2r_c \cos \theta$$



$$r = 2r_c \cos \theta$$

$$\dot{r} = 2r_c (-\sin \theta) \dot{\theta}$$

$$\ddot{r} = 2r_c (-\sin \theta) \ddot{\theta} + 2r_c (-\cos \theta) \dot{\theta}^2$$



$$\begin{aligned} \sum F_r &= m a_r \\ \sum F_\theta &= m a_\theta \end{aligned}$$

$$m a_r = ? \quad m a_\theta = ?$$

$$m a_r = m (\ddot{r} - r \dot{\theta}^2)$$

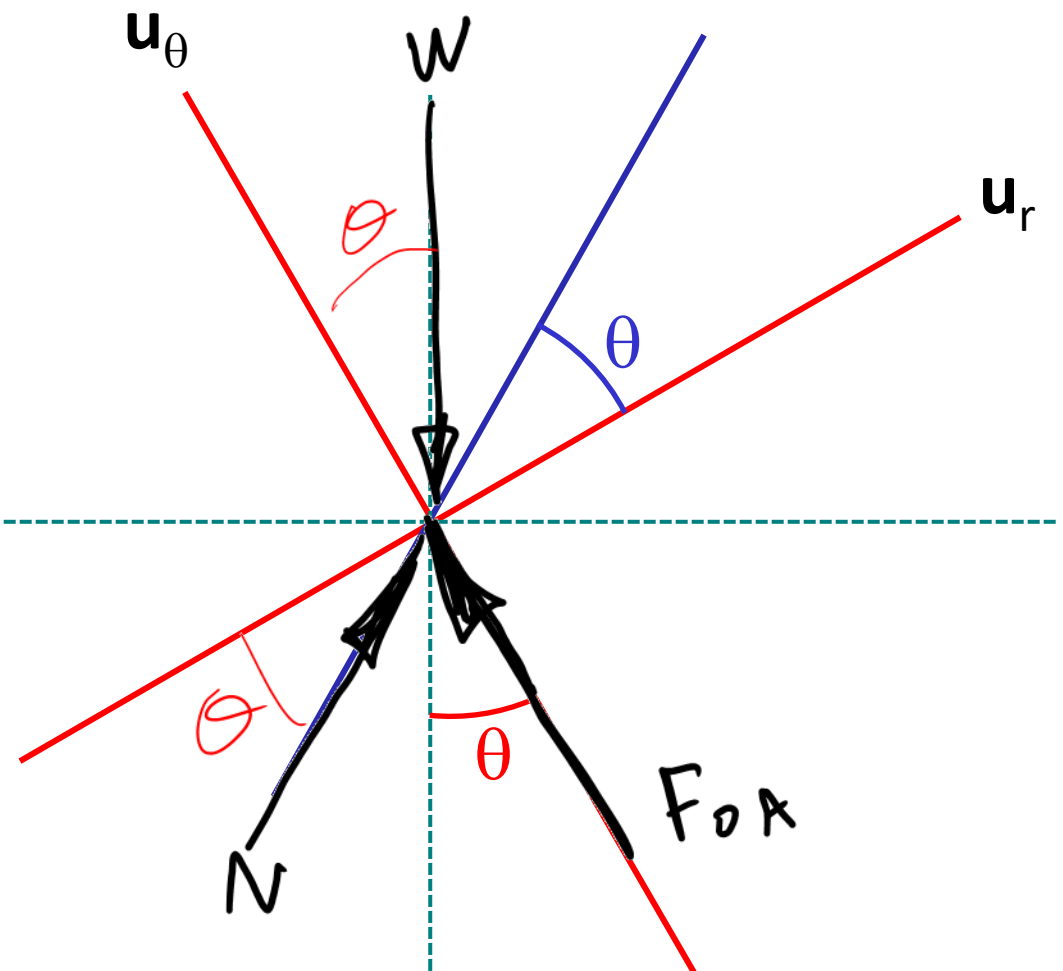
$$m a_\theta = m (r \ddot{\theta} + 2 \dot{r} \dot{\theta})$$

$$\dot{\theta} = 0.4 \text{ rad/s}$$

$$\ddot{\theta} = 0.8 \text{ rad/s}^2$$

$$\ddot{r} = ?$$

$$\dot{r} = ?$$



$$\sum F_r = N \cos \theta - W \sin \theta$$

$$\sum F_\theta = F_{OA} + N \sin \theta - W \cos \theta$$

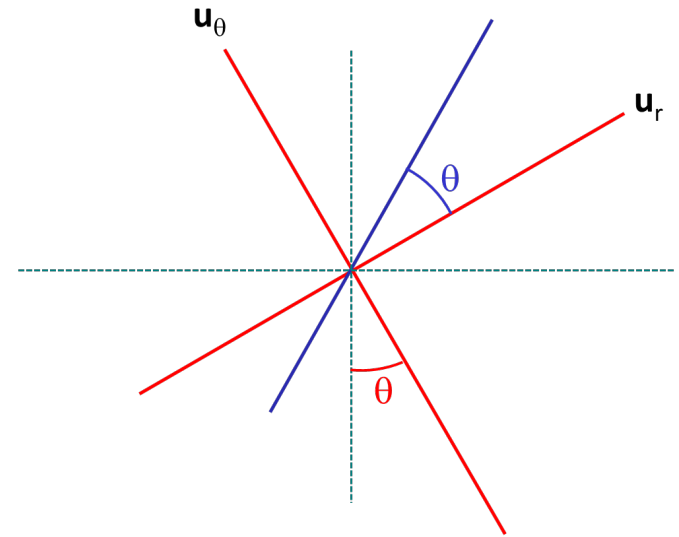
$$N = 0.279 \text{ lb}$$

$$F_{OA} = 0.300 \text{ lb}$$

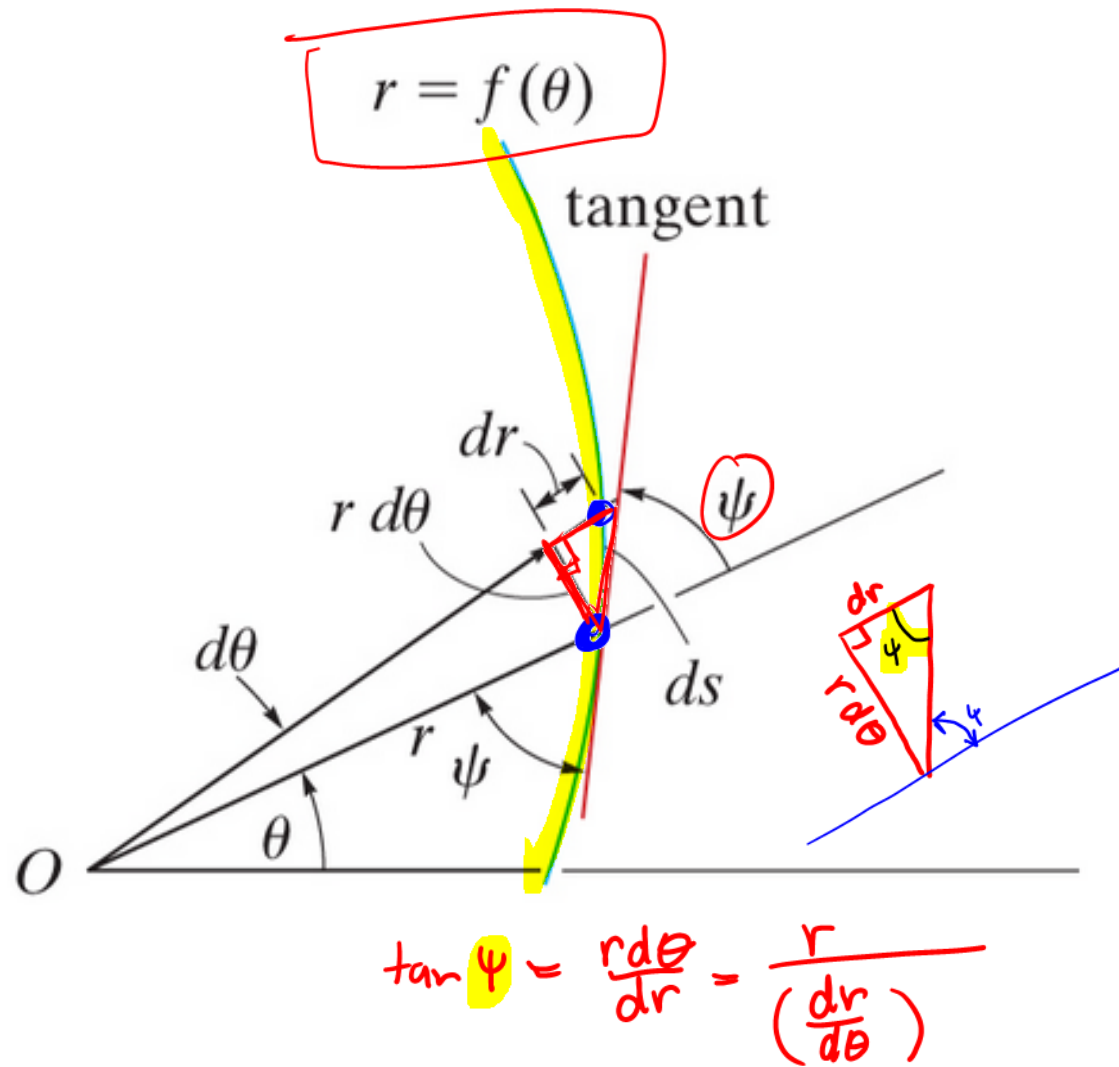
compute $\ddot{r}$, $\ddot{\theta}$ and do the algebra and you get this

$$N \cos \theta - W \sin \theta = m(\ddot{r} - r\dot{\theta}^2)$$

$$F_{OA} + N \sin \theta - W \cos \theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$



Tangential and Normal Directions



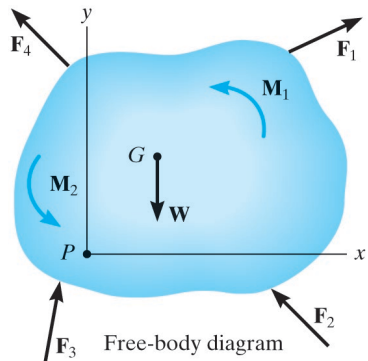
$$\tan \psi = \frac{r}{dr/d\theta}$$

For a particle $\vec{F} = m\vec{a}$

Equations of Motion for a Rigid Body

$$\Sigma \mathbf{F} = m\mathbf{a}_G$$

$$\Sigma M_G = I_G \alpha$$



m is the mass

I_G is the moment of inertia about the center of mass.

$\Sigma \mathbf{F}$ is the sum of the forces

ΣM_G is the sum of the moments of the external forces

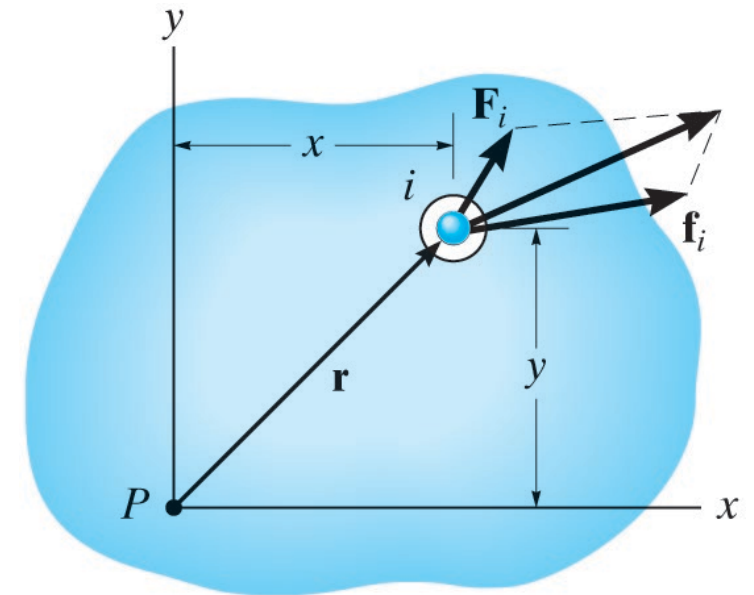
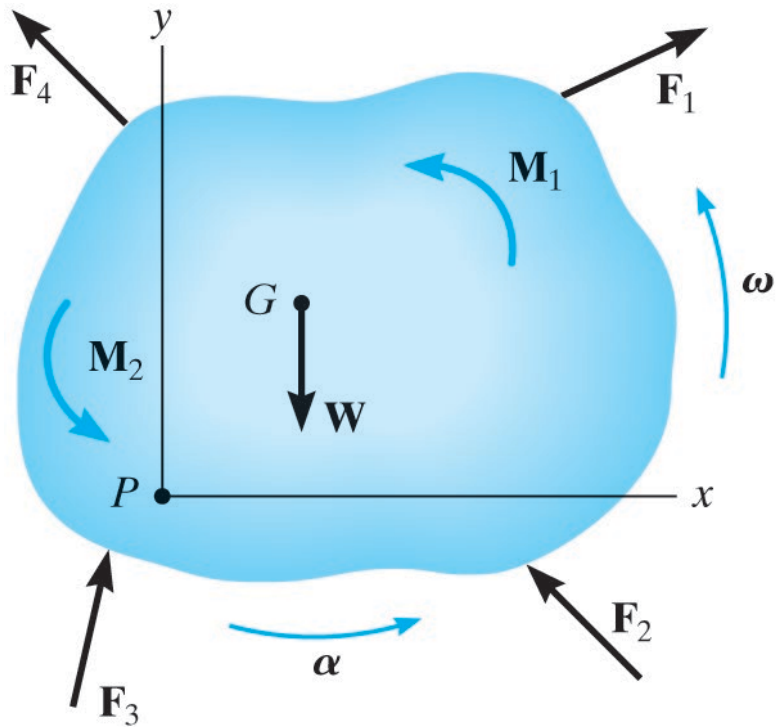
$\mathbf{a}_G$ is the acceleration of the center of mass

α is the ^{angular} ~~acceleration of the center of mass~~

Where do we get these equations?

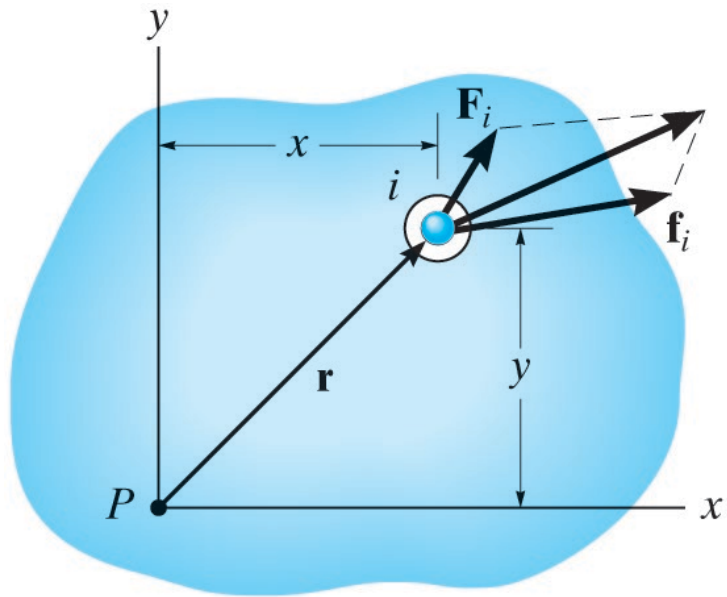
Remember, Newton's laws describe particles
not rigid bodies!

Extending Newton's 2nd Law to Rigid Bodies



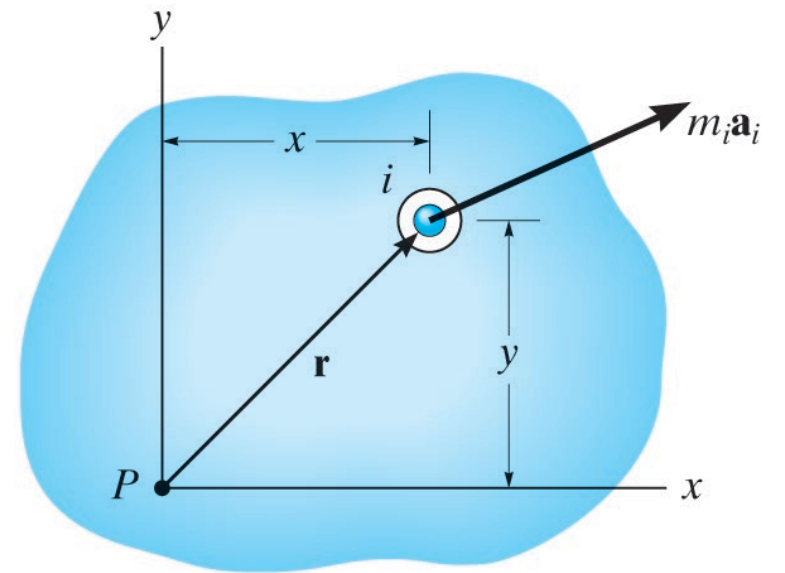
Consider the body to be a collection of particles!

For a **particle**: $\vec{F} = m\vec{a}$



Particle free-body diagram

=



Particle kinetic diagram

$$\mathbf{F}_i + \mathbf{f}_i = m_i \mathbf{a}_i$$

For a particle:

$$\mathbf{F}_i + \mathbf{f}_i = m_i \mathbf{a}_i$$

Sum over all particles:

$$\Sigma \mathbf{F}_i + \Sigma \mathbf{f}_i = \Sigma m_i \mathbf{a}_i$$

The sum of the internal forces is zero because the forces between particles cancel in pairs.

$$\Sigma \mathbf{F}_i = \Sigma m_i \mathbf{a}_i$$

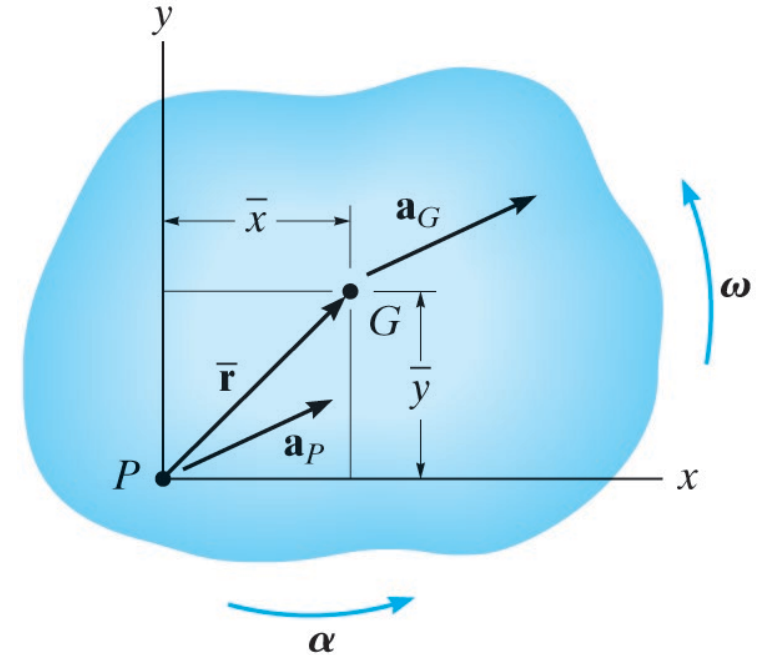
Consider the location of the center of mass G

$$\mathbf{r}_G = \frac{\Sigma m_i \mathbf{r}_i}{\Sigma m_i} \longrightarrow m \mathbf{r}_G = \Sigma m_i \mathbf{r}_i$$

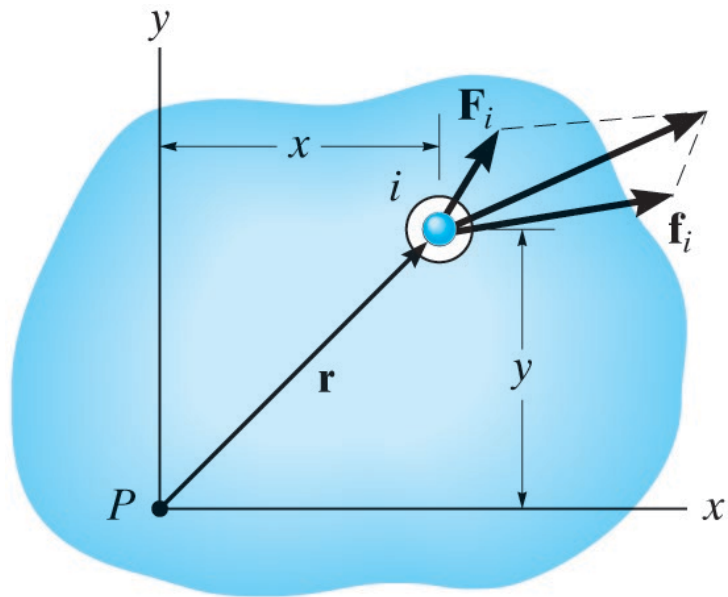
Differentiating twice gives:

$$m \mathbf{a}_G = \Sigma m_i \mathbf{a}_i$$

Force Equation for a Rigid Body

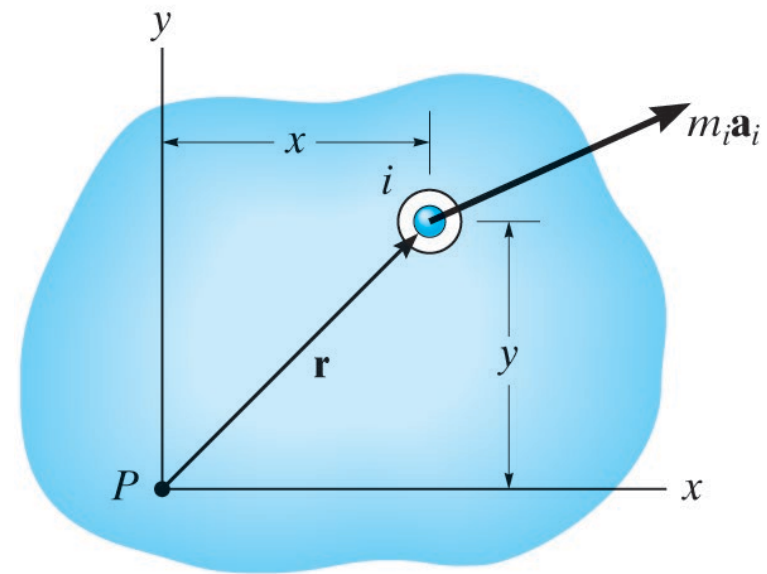


$$\Sigma \mathbf{F} = m \mathbf{a}_G$$



Particle free-body diagram

=



Particle kinetic diagram

$$\mathbf{r} \times \mathbf{F}_i + \mathbf{r} \times \mathbf{f}_i = \mathbf{r} \times m_i \mathbf{a}_i$$

$$(\mathbf{M}_P)_i = \mathbf{r} \times m_i \mathbf{a}_i$$

$$(\mathbf{M}_P)_i = \mathbf{r} \times m_i \mathbf{a}_i$$

Moment Equation for a Rigid Body

Express $\mathbf{a}_i$ relative to point P

$$\begin{aligned} (\mathbf{M}_P)_i &= m_i \mathbf{r} \times (\mathbf{a}_P + \boldsymbol{\alpha} \times \mathbf{r} - \omega^2 \mathbf{r}) \\ &= m_i [\mathbf{r} \times \mathbf{a}_P + \mathbf{r} \times (\boldsymbol{\alpha} \times \mathbf{r}) - \omega^2 (\mathbf{r} \times \mathbf{r})] \end{aligned}$$

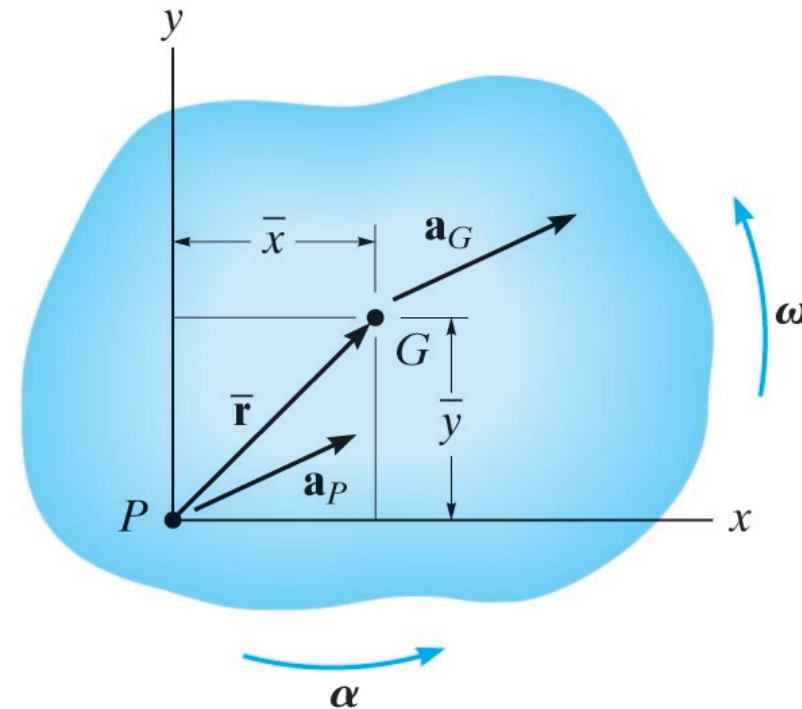
For a rigid body, the accelerations of the particles are not independent! We can relate the motion of individual particles to the motion of the body as a whole.

Plug in $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ and keep the non-zero part, which is $\mathbf{k}$:

$$\begin{aligned} (M_P)_i \mathbf{k} &= m_i \{ (x\mathbf{i} + y\mathbf{j}) \times [(a_P)_x \mathbf{i} + (a_P)_y \mathbf{j}] \\ &\quad + (x\mathbf{i} + y\mathbf{j}) \times [\boldsymbol{\alpha} \mathbf{k} \times (x\mathbf{i} + y\mathbf{j})] \} \end{aligned}$$

Expand the cross products:

$$\begin{aligned} (M_P)_i \mathbf{k} &= m_i [-y(a_P)_x + x(a_P)_y + \underbrace{\alpha x^2 + \alpha y^2}_{r^2 = x^2 + y^2}] \mathbf{k} \\ \sum (M_P)_i &= m_i [-y(a_P)_x + x(a_P)_y + \alpha r^2] \end{aligned}$$



$$\zeta (M_P)_i = m_i [-\bar{y}(a_P)_x + \bar{x}(a_P)_y + \alpha r^2]$$

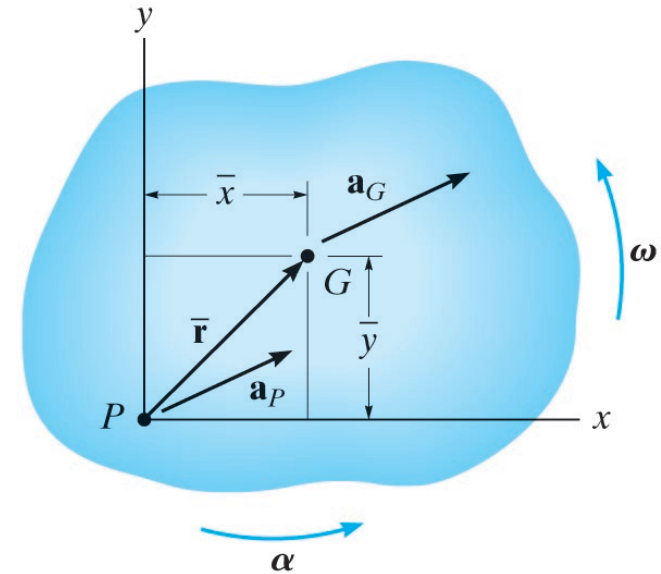
Moment Equation for a Rigid Body

Change from m_i to dm and integrate over the body:

$$\zeta \Sigma M_P = -\left(\int_m y dm\right)(a_P)_x + \left(\int_m x dm\right)(a_P)_y + \left(\int_m r^2 dm\right)\alpha$$

This gives:

$$\zeta \Sigma M_P = -\bar{y}m(a_P)_x + \bar{x}m(a_P)_y + I_P\alpha$$



If P is selected so that $P = G$,
we get a much simpler result:

$$\Sigma M_G = I_G \alpha$$

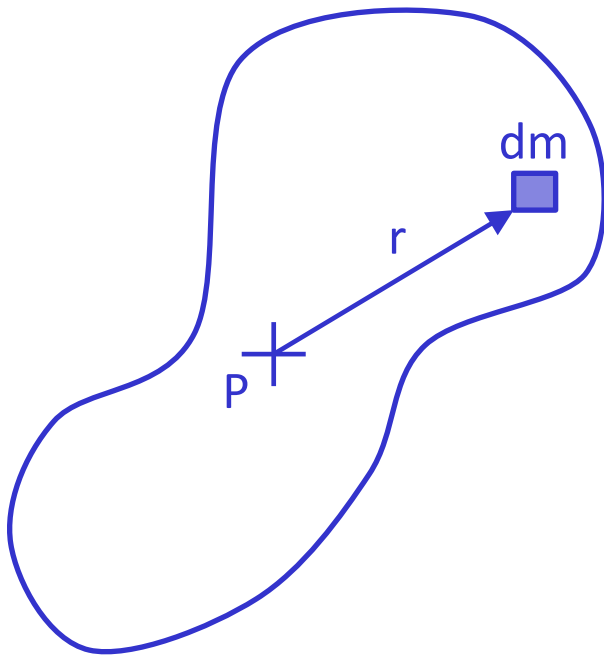
I_G is the moment of inertia about the center of mass.

ΣM_G is the sum of the external moments because the internal moments are due to forces between the particles which cancel in pairs.

Returning from the Derivations

Moment of Inertia

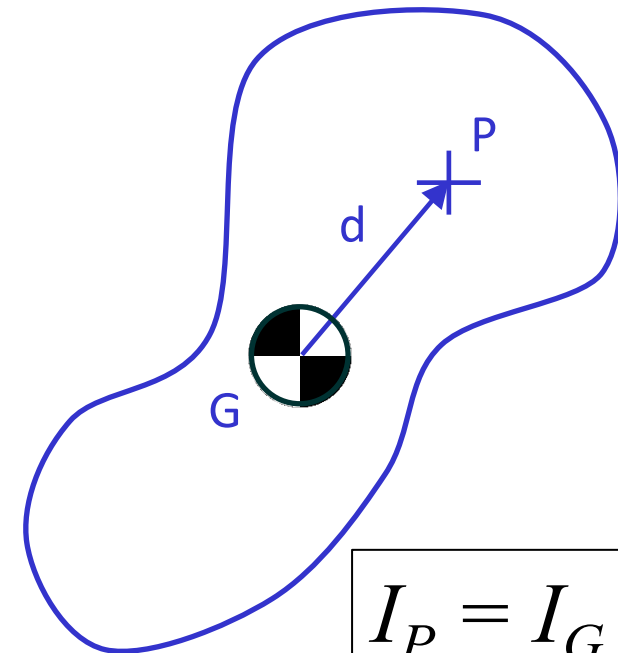
$$I_P = \int_m r^2 dm$$



Radius of gyration

$$I = mk^2 \quad \text{or} \quad k = \sqrt{\frac{I}{m}}$$

Parallel Axis Theorem



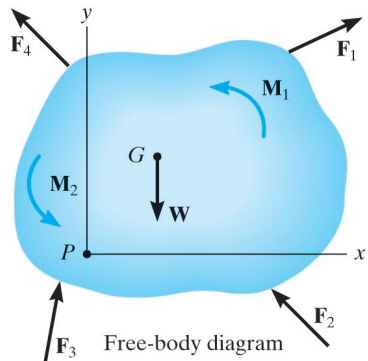
$$I_P = I_G + md^2$$

Only works relative to center of mass.

Equations of Motion for a Rigid Body

$$\Sigma \mathbf{F} = m \mathbf{a}_G$$

$$\Sigma M_G = I_G \alpha$$



m is the mass

I_G is the moment of inertia about the center of mass.

$\Sigma \mathbf{F}$ is the sum of the forces

ΣM_G is the sum of the moments of the external forces

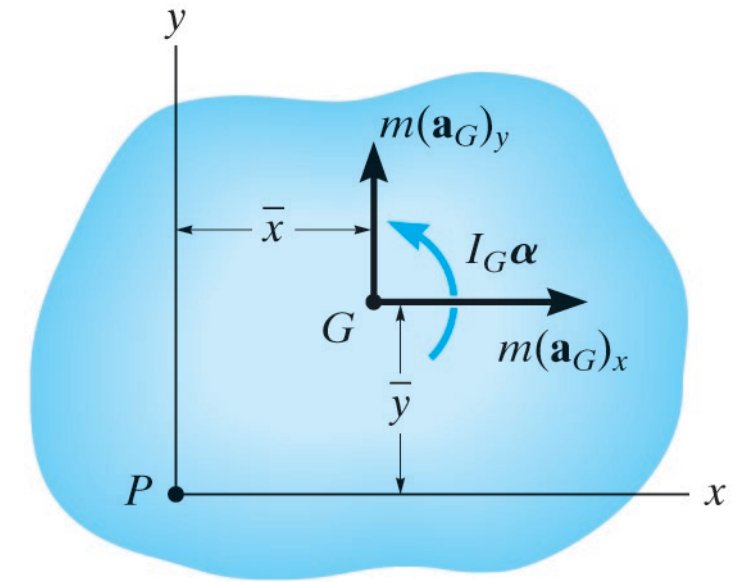
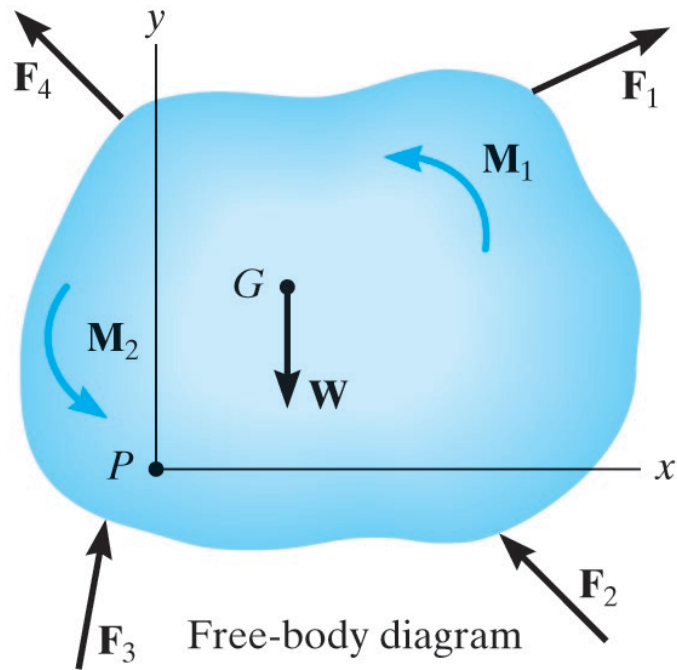
$\mathbf{a}_G$ is the acceleration of the center of mass

α is the acceleration of the center of mass

Equations of Motion for a Rigid Body

Planar Motion

$$\begin{aligned}\Sigma F_x &= m(a_G)_x \\ \Sigma F_y &= m(a_G)_y \\ \Sigma M_G &= I_G \alpha\end{aligned}$$

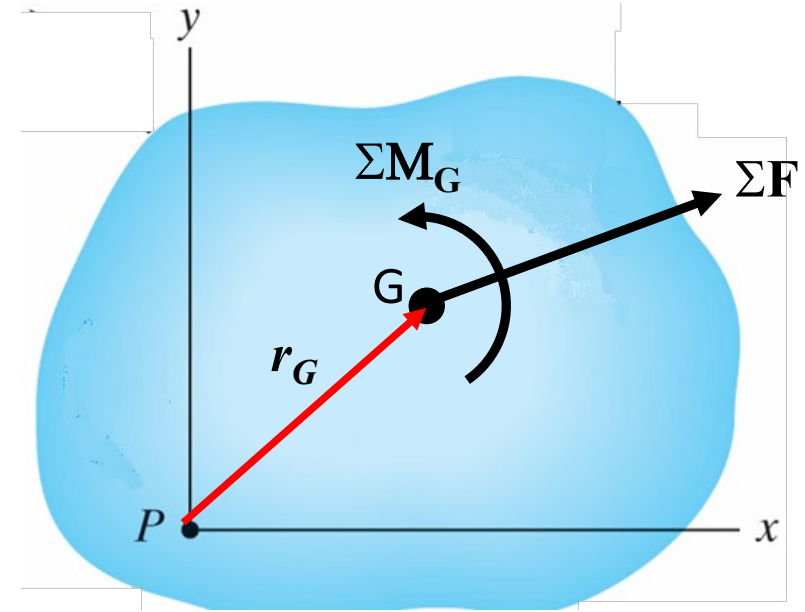
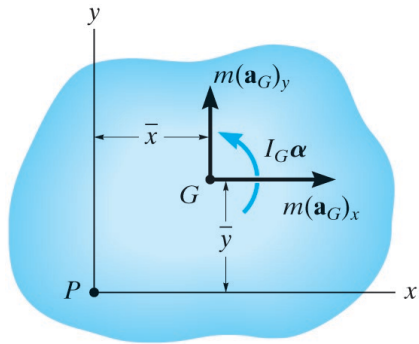


Equations of Motion for a Rigid Body

Planar Motion

Moments taken about center of mass.

$$\begin{aligned}\Sigma F_x &= m(a_G)_x \\ \Sigma F_y &= m(a_G)_y \\ \Sigma M_G &= I_G \alpha\end{aligned}$$



$$\Sigma M_P = \Sigma M_G + r_G \times (\Sigma F)$$

$$\Sigma M_P = I_G \alpha + r_G \times (m a_G)$$

Equations of Motion for a Rigid Body

Pure Translation in the Plane:

$$\begin{aligned}\Sigma F_x &= m(a_G)_x \\ \Sigma F_y &= m(a_G)_y \\ \Sigma M_G &= 0\end{aligned}$$

